

# Vincent (Dam Duc Huy) Ngo

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## EDUCATION

### University of Alberta

*B.Sc. Computer Engineering, Co-op - GPA: 3.9/4.0*

Edmonton, AB

*Expected 2029*

## ENGINEERING EXPERIENCE

### Electrical and Control Engineering Intern

May 2026 – Aug. 2026

*Red Deer Polytechnic - CIM-TAC / Energy Innovation Centre*

*Red Deer, AB*

- Architected dual-core **FreeRTOS** firmware in C/C++ on ESP32-S3 with bare-metal C on PIC18F24Q10 for a field-deployed sensing platform; reduced average current draw **70.4% (85.71 mA to 25.33 mA)** using GPIO ISRs, task notifications, peripheral power gating, and watchdog-based sleep scheduling.
- Diagnosed a FreeRTOS soft-lock caused by queue blocking during E-paper task teardown and implemented a deterministic queue-bypass shutdown path; replaced static sensing thresholds with a rate-of-change state machine using a **300 s SRAM ring buffer** and EMA filtering.
- Engineered an in-circuit **Spy-Bi-Wire ROM BSL flasher** in C++/FreeRTOS on ESP32-C6 for MSP430FR6043, streaming CRC-16 validated 240-byte payloads to 20-bit FRAM at 115,200 baud and eliminating parasitic back-powering through defensive High-Z GPIO sequencing.
- Migrated embedded BLE firmware from Bluedroid to NimBLE, reducing static binary footprint by approximately **600 KB (over 35%)** and enabling an 8 MB dual-image OTA partition layout with automated firmware staging and cross-platform build tooling.
- Designed firmware and hardware for a **65 mm x 18 mm, 4-layer mixed-signal PCB** based on ESP32-S3, integrating IMU and force sensing with low-noise power rails and streaming 180-byte batched telemetry at 50 Hz over BLE to a companion application.
- Implemented a **100 Hz FreeRTOS PID control task** on ESP32-S3 for a 4.5 kW servo system, interfacing isolated analog command/feedback, RS-485/Modbus, and 4-20 mA instrumentation while supporting safe hardware bring-up and troubleshooting.

### Flight Software & Embedded Systems Team Member

Sept. 2024 – Apr. 2025

*AlbertaSat - Ex-Alta 3 CubeSat*

*Edmonton, AB*

- Contributed 7+ months to a 3U CubeSat OBC platform spanning **embedded Linux (Yocto/PetaLinux), RTOS, BIOS/boot configuration, Rust flight software, and FPGA-based digital logic** for mission operations in LEO.
- Developed asynchronous Rust flight-software handlers, ground-station CLI tooling, and automation scripts; supported fault-tolerant boot, watchdog, and telemetry workflows across the spacecraft computing stack.
- Performed hardware-in-the-loop integration and ISO 8 clean-room qualification, debugging UART, SPI, I2C, and CAN interfaces with oscilloscopes and logic analyzers and authoring version-controlled verification documentation.

## EMBEDDED SYSTEMS PROJECTS

### Bare-Metal Real-Time Audio Spectrum Visualizer | STM32, C, CMSIS-DSP, DMA, PWM

2025

- Built a register-level STM32F103C8 firmware pipeline using ADC, DMA, timer interrupts, and a **1024-point CMSIS-DSP FFT**; profiled and reduced processing from 28 ms to **15 ms**, achieving **17 ms end-to-end latency** within a 20 KB SRAM budget.
- Generated nanosecond-accurate 800 kHz WS2812B waveforms using timer PWM while maintaining non-blocking audio acquisition; validated DMA throughput and timing with GDB, a logic analyzer, and oscilloscope.

## EMBEDDED SYSTEMS PROJECTS

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- NEST Wearable Health Monitor** | *ESP32, ESP-IDF, FreeRTOS, KiCad* 2025
- Architected a dual-core ESP-IDF/FreeRTOS firmware stack separating high-rate PPG/IMU acquisition from GUI and Wi-Fi tasks, using queues, semaphores, mutexes, and SPI DMA to prevent display-induced task starvation.
  - Designed and assembled a compact 2-layer mixed-signal PCB integrating MAX30102, MPU6500, ENS160, and AHT21 sensors; validated 3.3 V rails and I2C/SPI signal integrity and implemented power-aware sensor duty cycling.
- Automated Embedded Build & Deployment Pipeline** | *Bash, PowerShell, WSL2, XC8* 2026
- Engineered a headless PowerShell/Bash build-and-flash pipeline around `arduino-cli` and native Linux `xc8-cc`, standardizing multi-target builds, flash layouts, and OTA/in-circuit firmware staging while cutting deployment overhead by **over 60%**.
  - Resolved Windows/WSL compiler interoperability and path-translation failures by wrapping vendor toolchains in POSIX-compliant scripts and automating environment configuration for repeatable firmware builds.
- High-Throughput Serial & BLE Telemetry Engine** | *C#, .NET 8, System.IO.Pipelines* 2026
- Built a multithreaded telemetry engine using `System.IO.Pipelines` and thread-safe channels for resilient serial/BLE framing, checksum validation, mixed-endian decoding, and sustained diagnostic logging without UI-thread stalls.
  - Implemented a zero-allocation byte-stream framing reader and bounded background-worker pipeline, eliminating buffer backpressure stalls and packet loss during high-throughput hardware diagnostics.

## TECHNICAL SKILLS

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**Programming:** C, C++, Embedded C, Rust, Python, C#, Bash, PowerShell; multithreaded systems, CLI tools, Linux/POSIX environments

**Embedded & Real-Time:** FreeRTOS, ESP-IDF, bare-metal ARM/CMSIS, task scheduling, queues, mutexes, task notifications, critical sections, GPIO, ADC, DMA, timers, ISRs, peripheral drivers

**MCUs & Platforms:** STM32 (ARM Cortex-M), ESP32-S3/C6, MSP430FR6043, PIC18F24Q10, FPGA/SoC platforms

**Embedded Linux & Toolchains:** Yocto Project, PetaLinux, Buildroot, WSL2, GNU Arm, Microchip XC8, TI CCS/MSP430 CGT, cross-compilation, boot/firmware utilities

**Protocols & Connectivity:** UART/USART, SPI, I2C, CAN, BLE, Wi-Fi, TCP/IP, RS-485/Modbus RTU, 4-20 mA, JTAG/SWD, Spy-Bi-Wire

**Debugging & Test:** Git, GDB, oscilloscopes, logic analyzers, DMMs, protocol analysis, Nordic PPK2, hardware bring-up, firmware debugging, hardware-software integration

**Hardware & DSP:** KiCad, Altium Designer, mixed-signal PCB design, SMT assembly/rework, CMSIS-DSP, FFT, real-time signal processing, VHDL, Verilog, Vivado

## AWARDS & SCHOLARSHIPS

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- IEEE Northern Canada Section Scholarship** 2026
- The Albert Julian Hrapko Memorial Scholarship in Computer Engineering** 2025
- The Robert Stollery Scholarship in Engineering** 2024–2025
- Third Place - Natignite 2025 Hackathon** 2025